

Rate of turn, turn radius and the steering linkage — Northstar 76, 2026 season

26 September 2026. Every number is measured from the boat's own logs. Where the logs cannot support a measurement it is left out rather than estimated, and the coverage is stated beside each figure.

1. The three numbers, and why they differ

Rate of turn — heading change over the **6 seconds centred on the moment the apparent wind crosses the bow**, divided by the span actually measured. The average through the fastest part of the turn.

Turn radius — **speed ÷ yaw rate** over that same window. The radius the boat actually turned at. This is the operating point any steering geometry has to be correct for, and unlike the geometry it can be measured without knowing anything about the boat.

Max rotation — the single fastest heading change between two consecutive samples. A *peak*, which rises with the log's own sampling noise.

Compared **on the same manoeuvres**, max rotation reads higher in **94 of 98 tacks** and **44 of 44 gybes**, by a median of **1.00 °/s** and **1.19 °/s**. That gap is the sampling, not the sailing. A target set from max rotation would be a target nobody is actually sailing.

Timing the window on the apparent wind rather than on the manoeuvre detector matters: on 12 September the crossing lands a consistent **+2.7 s after** the detected moment, so timing from the detector would have measured a different part of every turn.

What the season can carry

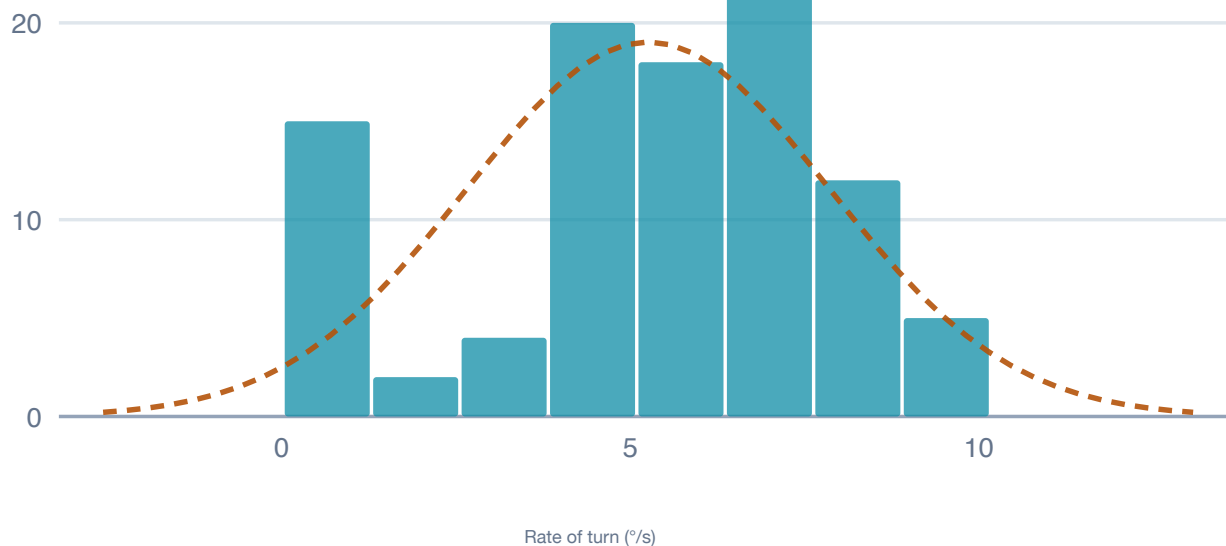
	tacks	gybes
Rate of turn (±3 s window)	98 of 587, on 8 of 23 days	44 of 230, on 6 of 22 days
Turn radius	81	42
Max rotation	546	220

Rate of turn needs a log sampled at **3 s or finer**; half the season is logged at 6.09 s, and a six-second window sampled every six seconds is not a measurement. Turn radius needs the rate *and* a real turn under it — **$R = v/\omega$ runs away as ω goes to zero**, so it is only computed above 2 °/s. Max rotation survives a coarse log, which is why it covers five times as many manoeuvres, and why it should be read as a ceiling rather than a performance.

2. Tacks

Rate of turn across 98 tacks

■ Rate of turn · 98



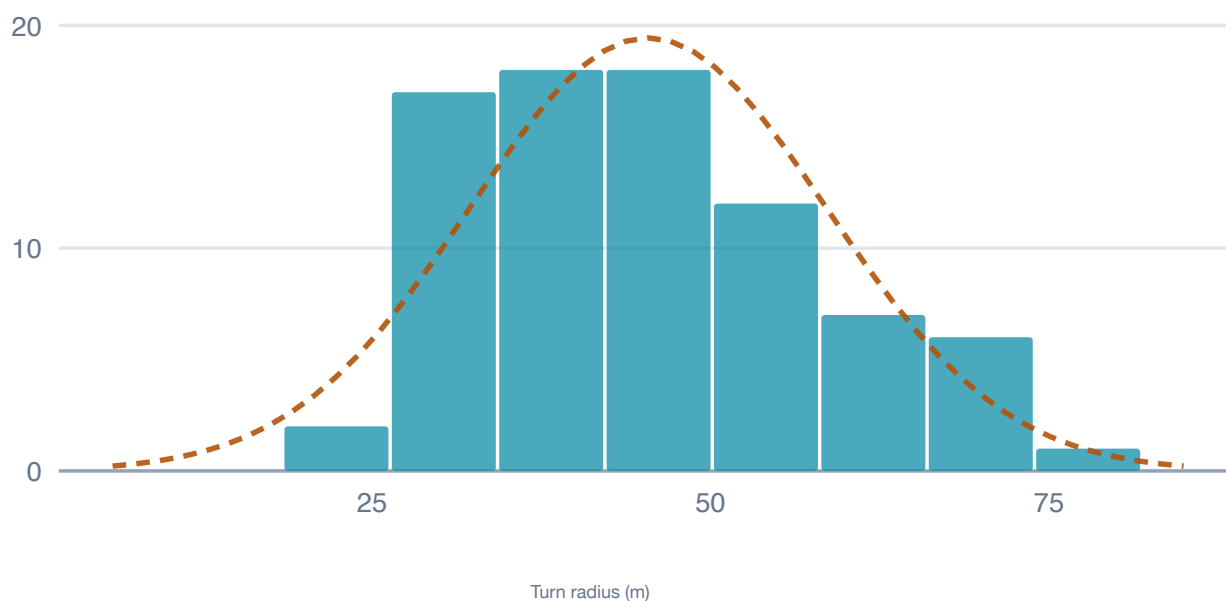
Rate of turn through 98 tacks. Bars are counts; the dashed line is the normal fitted to the same mean and spread, drawn so the difference from it can be seen.

It is not a bell. A hump at 5–7 °/s and a tail of slow ones running to nearly zero. The mean of **5.25 °/s** sits below the median of **5.73** because that tail drags it there; the typical tack is the median.

The bottom bar is not slow tacking. **Fifteen of the ninety-eight** sit near zero, and a tack that turns at 0.3 °/s is something recorded as a tack that was not one, or a stretch where the heading data is bad. They are excluded from the radius for that reason.

Turn radius across 81 tacks

■ Turn radius · 81

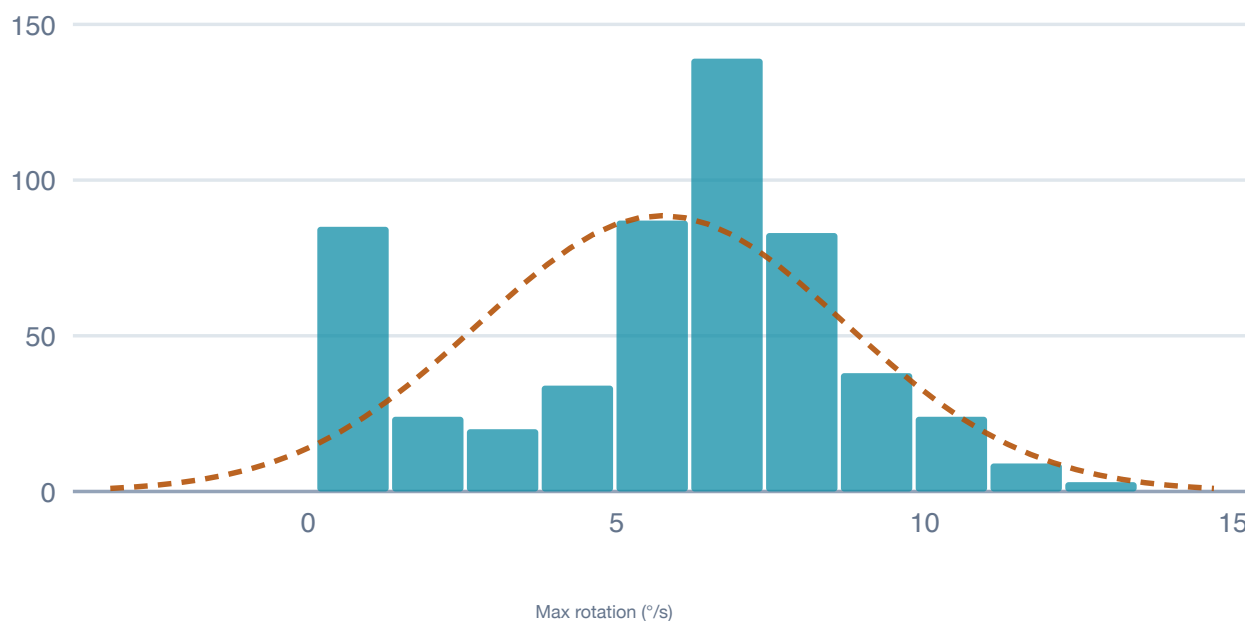


Turn radius through 81 tacks, in metres.

The boat tacks in about 43 m — median, against a hull length of 23.16 m, so **1.86 boat lengths**, and tightly grouped: 18 m to 82 m with a spread of 13 m.

Max rotation across 546 tacks

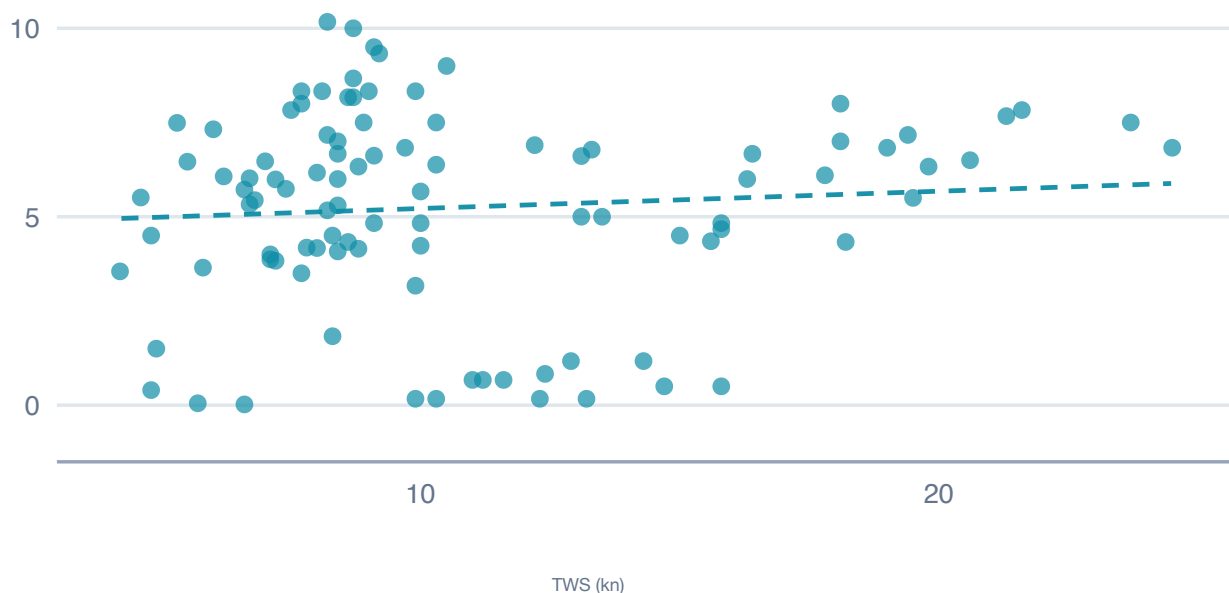
■ Max rotation · 546



Max rotation through 546 tacks — five times the sample, because a coarse log still supports it.

Rate of turn vs TWS (°/s)

● Tacks · 98 · R^2 0.01

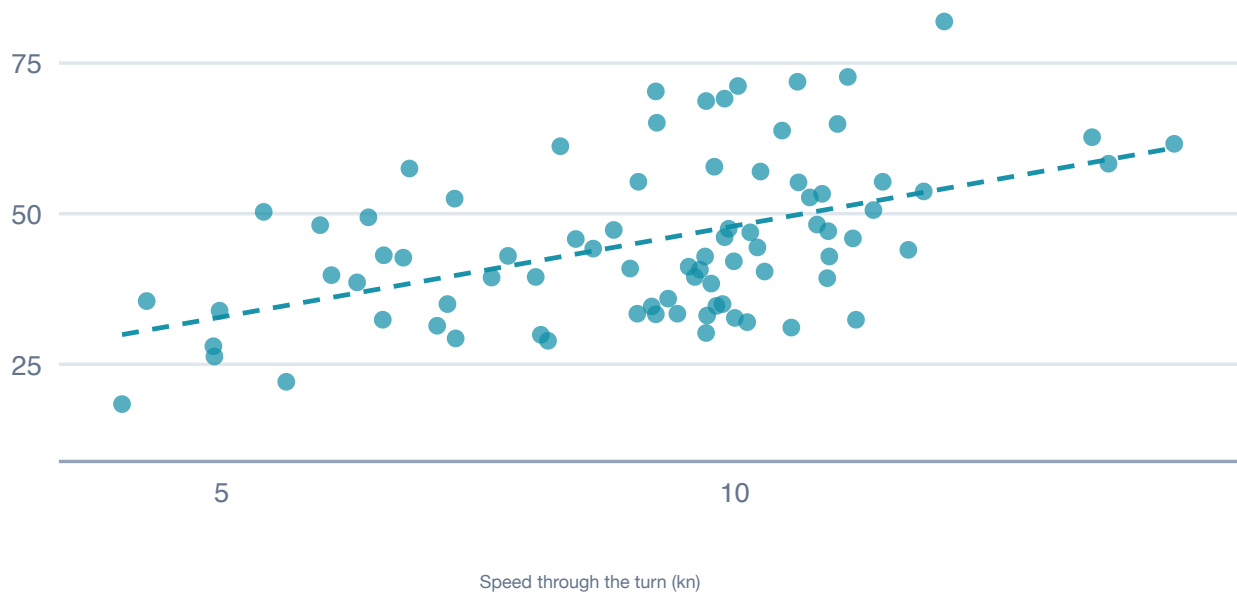


Rate of turn against true wind speed, 98 tacks, with a least-squares line.

Rate of turn does not depend on the breeze. R^2 is **0.01** across 7–20 knots. Whatever governs how fast this boat comes through the wind, it is not wind strength — so it is worth looking at what the crew does rather than at conditions.

Turn radius vs Speed through the turn (m)

● Tacks · 81 · R^2 0.24

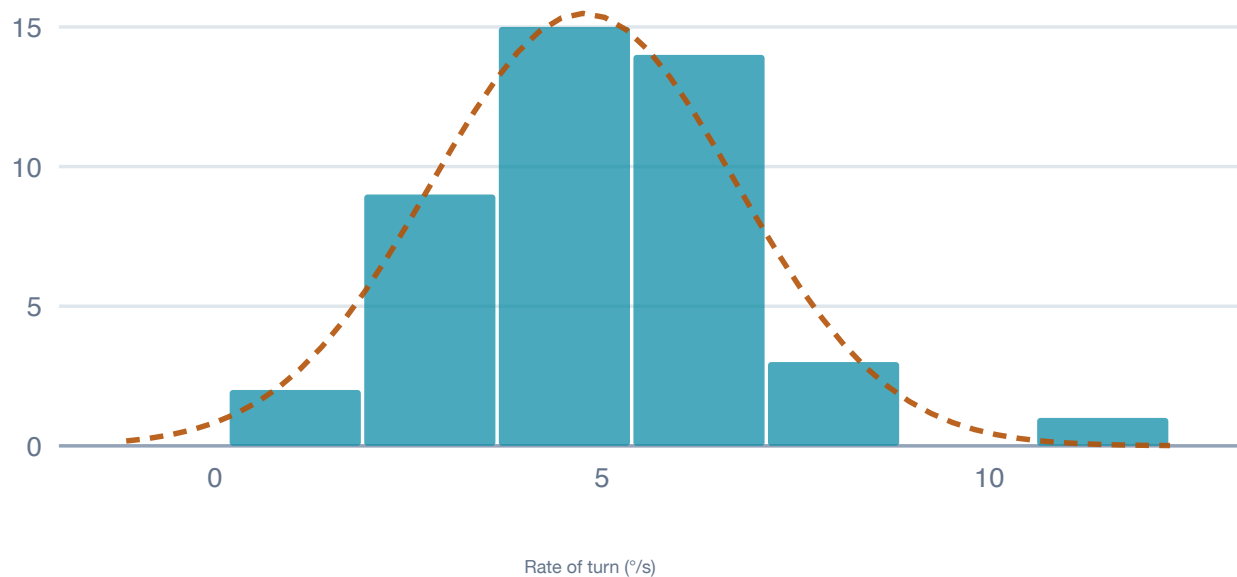


Turn radius against the speed carried through the turn.

3. Gybes

Rate of turn across 44 gybes

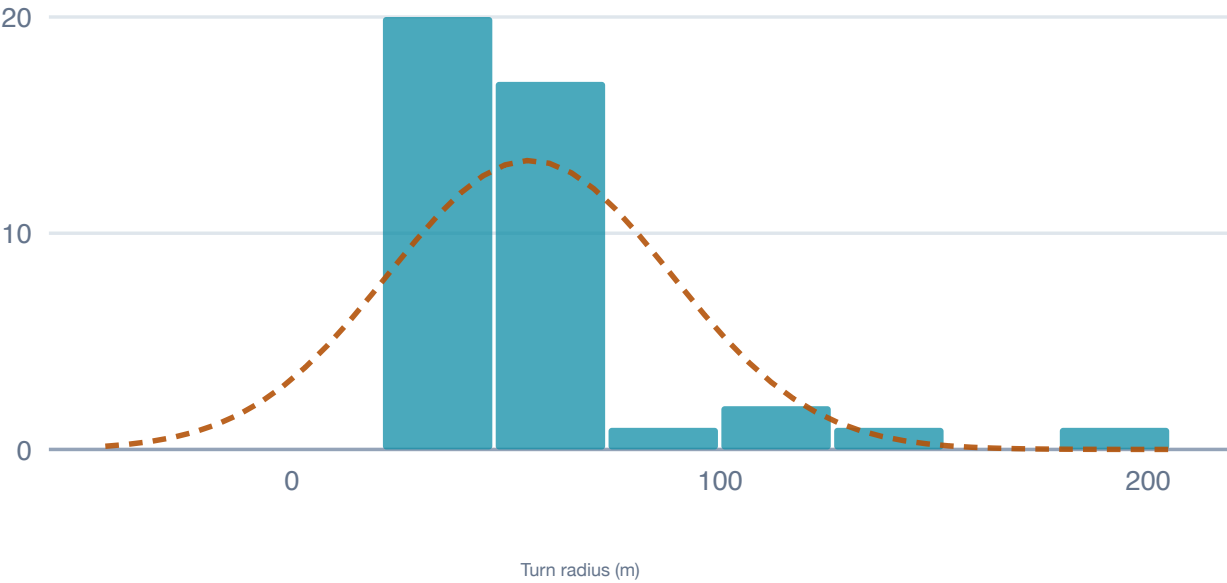
■ Rate of turn · 44



Rate of turn through 44 gybes.

Turn radius across 42 gybes

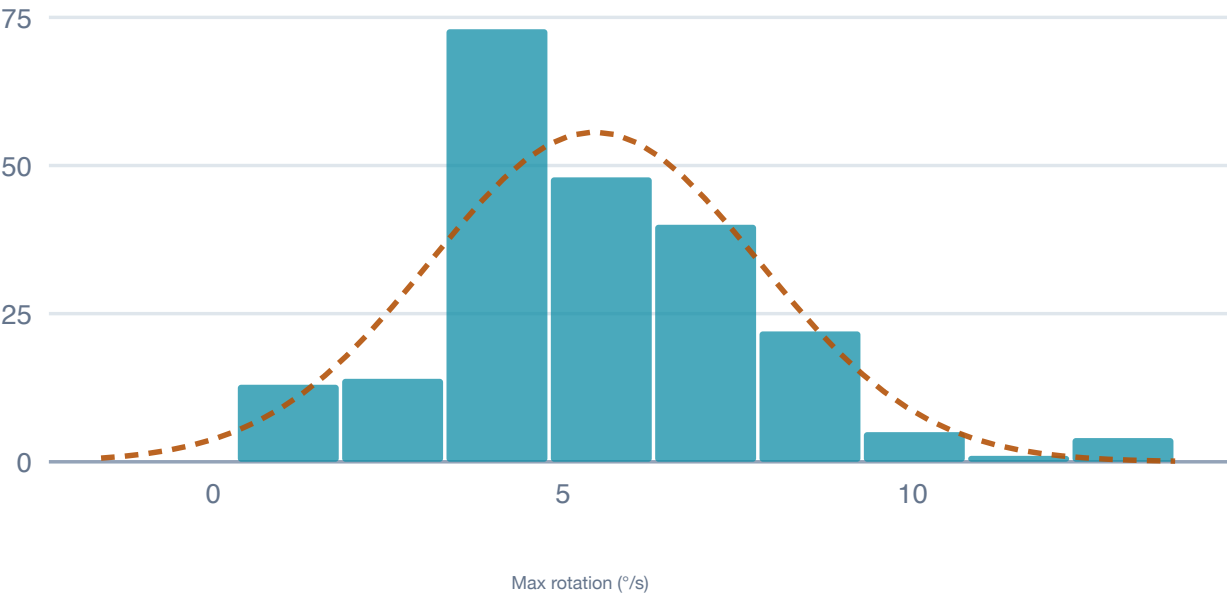
■ Turn radius · 42



Turn radius through 42 gybes.

Max rotation across 220 gybes

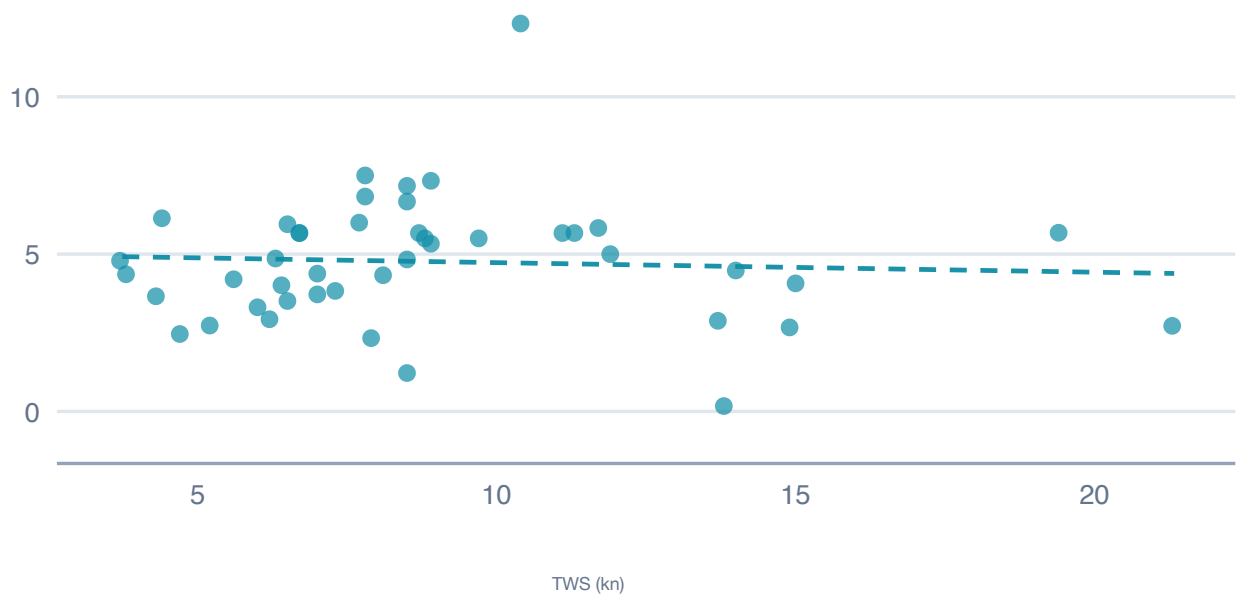
■ Max rotation · 220



Max rotation through 220 gybes.

Rate of turn vs TWS (°/s)

● Gybes · 44 · R² 0.00



Rate of turn against true wind speed, 44 gybes.

Gybes turn **slower and wider** than tacks — 4.81 °/s against 5.73, and a radius of **49.6 m (2.15 boat lengths)** against 43.0 — but the rate is more tightly grouped (spread 1.97 against 2.61). A gybe is a smaller heading change made under more control, and it is the more repeatable manoeuvre on this boat. The radius spread is the wider of the two (33 m against 13 m), driven by a handful of very wide ones out to 205 m.

4. Summary

- The typical tack turns at **5.7 °/s** in **43 m**; the typical gybe at **4.8 °/s** in **50 m**. Medians, from the ± 3 s window.
- The boat turns in **roughly two boat lengths**, and tacks are the tighter of the two manoeuvres.
- The tack rate distribution is **not normal** — a hump with a tail of slow ones. The tail is where the time goes, not the average.
- **Rate of turn is independent of wind speed** (R² 0.01). If it is to be improved, the variable is not the breeze.
- **Fifteen of the ninety-eight rated tacks look like bad records rather than bad tacks** and should be checked before any mean is quoted onward.
- Max rotation reads higher on the same manoeuvre almost every time. Use it for coverage, not as a target.

5. Steering geometry

The operating point, measured

This is what turn radius buys: the geometry only has to be right where the boat actually sails, and that is now known rather than assumed.

Treating the boat as a bicycle — a single effective steering angle at an effective distance **L** ahead of the turn centre — the helm carried through each turn gives

$$L = R \cdot \tan \delta$$

Across **68 tacks** that lands at a median **12.6 m** (range 5.5–19.6), and across **36 gybes** at **11.7 m**. The two agree, which they need not have, and both are close to **half the hull length**.

That is an *effective* pivot distance, not a tape measurement: a hull generates side force along its whole length, so it will not equal the geometric distance from the rudder stock to the centre of lateral resistance. It is, however, the number the boat behaves as though it has — **worth comparing against the tape measurement rather than replaced by it**.

At the measured operating point — radius **43 m**, L **12.6 m** — the helm angle implied is **16.3°**, which matches the top of the measured steering range.

The linkage, measured

Across **78,149 log rows** carrying both rudder angles, fitting starboard against port gives

$$\text{ruddS} = 1.0008 \times \text{ruddP} - 1.215 \quad (R^2 \ 0.932)$$

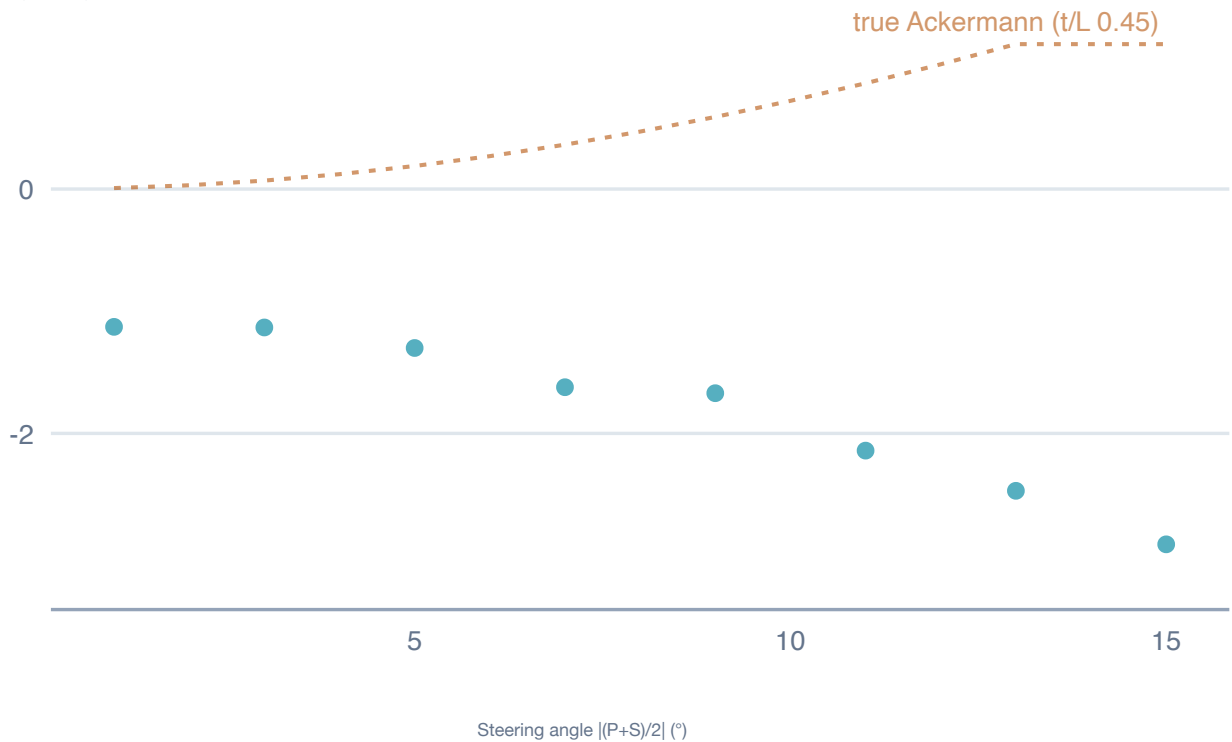
A slope of **1.0008** is **parallel steering**: within these sensors' resolution the two rudders turn by the same amount, with a **constant offset of about –1.2°** rather than a progressive differential.

The differential does drift with steering angle — weakly, and the *wrong way*:

steering angle	rows	mean starboard – port
0–2°	43,058	-1.13°
2–4°	19,228	-1.13°
4–6°	6,996	-1.30°
6–8°	3,633	-1.62°
8–10°	1,990	-1.67°
10–12°	1,359	-2.14°
12–14°	1,042	-2.47°
14–16°	842	-2.91°

Rudder differential against steering angle (°)

● Measured (binned) · 78149



Measured differential (binned means) against steering angle, with the differential true Ackermann would require shown dashed. The measured curve runs the other way.

About **-0.09 to -0.15° per degree** of helm depending on turn direction, with R^2 of only **0.05 to 0.14** — a small systematic effect inside a lot of scatter.

Under true Ackermann the inner rudder turns **more** than the outer, making starboard – port **positive** in both directions. It is negative and grows more negative. **The boat is running slightly anti-Ackermann.**

The target, once the rudder separation is known

With **L = 12.6 m measured** and the operating radius of **43 m**, the only missing number is **t**, the separation between the rudder stocks. Inner and outer angles at that operating point:

t (m)	t / L	inner	outer	differential needed
4.0	0.317	17.08°	15.64°	+1.44°
4.5	0.357	17.18°	15.56°	+1.62°
5.0	0.397	17.28°	15.48°	+1.80°
5.5	0.437	17.38°	15.40°	+1.98°
6.0	0.476	17.48°	15.32°	+2.17°

Against a **measured -2.91°** at that helm. Whatever **t** turns out to be, the correction is of the order of **4.4° to 5.1° at full helm** — reversing the sign and adding the Ackermann. That is large enough to be worth getting right, and large enough that guessing **t** would have been irresponsible.

The one thing that must happen first

These are sensor readings, not linkage geometry. A constant **-1.2°** offset is indistinguishable, from the log alone, from one rudder's angle sensor being zeroed **1.2°** differently from the other. **Zero both sensors dockside against physically aligned rudders and record the offsets before anything mechanical is touched.** If the offset proves to be calibration, what remains is the **~0.1 °/°** drift, which is small — and the whole picture changes.

The linkage, solved for the measured boat

Two dimensions now measured rather than assumed: the rudder stocks are **2238 mm** either side of the centreline (**t = 4476 mm**) and the tiller arms are **349 mm**. With the operating point already measured — a **43 m** turn radius and an effective pivot distance of **12.6 m** — the linkage has one answer.

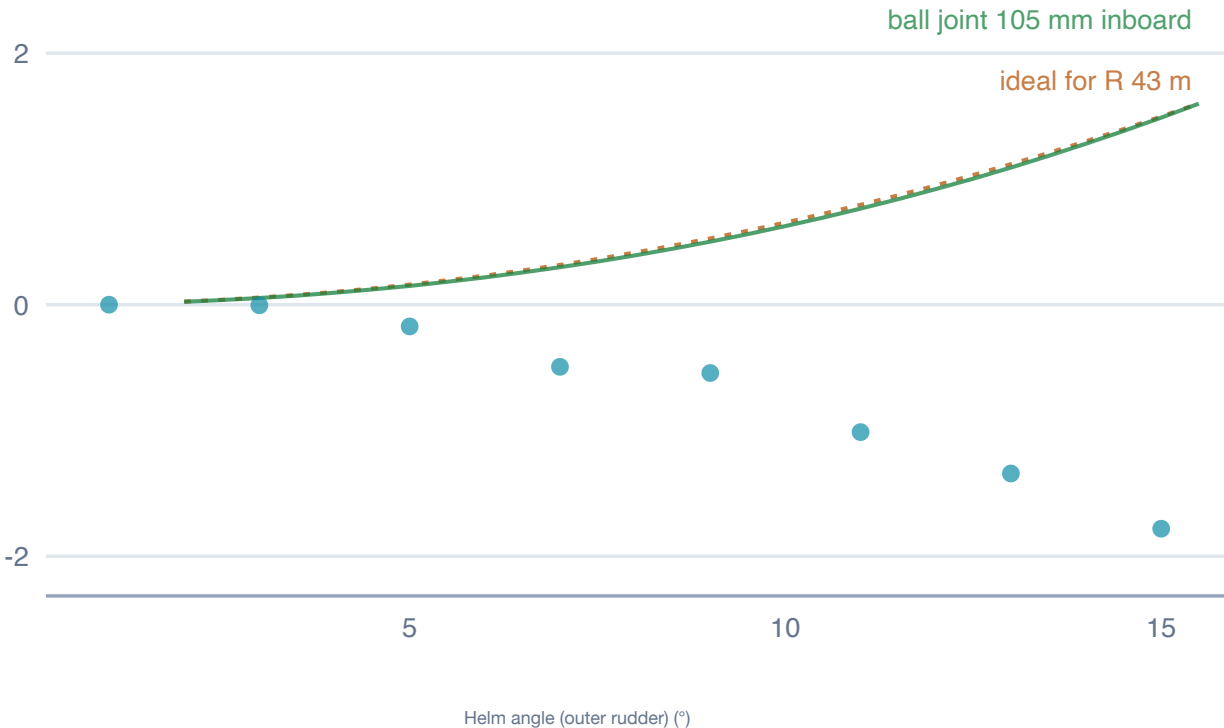
Rudder separation t	4476 mm (2 × 2238), fixed
Tiller length a	349 mm, fixed
t / L	0.355
Ideal differential at 43 m	+1.61° (inner 17.18°, outer 15.56°)
Arm angle required	17.5° inboard of the fore-and-aft line
Ball joint position	105 mm inboard, 333 mm aft of the rudder stock
Tie rod length	4266 mm — 210 mm shorter than the 4476 mm a parallel linkage needs

With the tiller length fixed at 349 mm, the adjustment is **where on that arm the tie-rod ball joint sits**: 105 mm off the fore-and-aft line, 333 mm aft. A new hole and a tie rod 210 mm shorter.

Note that the textbook rule — arms pointing at the pivot, $\text{atan}((t/2)/L) = 10.1^\circ$ — **under-delivers here**, giving only +0.84° against the +1.61° wanted. That rule is exact only for large radii and small helm; this boat turns tightly enough that the angle has to be solved for directly, which is what the table above is.

Rudder differential — measured now, and with the 349 mm tiller re-drilled (°)

● Measured now (offset removed) · 78149



What the re-drilled 349 mm tiller delivers across the steering range, against the ideal for a 43 m turn, with the measured behaviour (constant offset removed). The solid line sits on the ideal.

helm	delivered	ideal	error
5°	+0.15°	+0.16°	−0.01°
10°	+0.62°	+0.65°	−0.03°
15°	+1.49°	+1.50°	−0.01°
20°	+2.85°	+2.69°	+0.16°

How it holds across the turns the boat actually makes

The geometry is exact at one radius and approximate either side of it. Across the 18–82 m the boat really sails:

turn radius	helm	delivered	ideal	error
18 m	31.9°	+9.79°	+6.73°	+3.06°
25 m	24.8°	+4.81°	+4.14°	+0.67°
30 m	21.3°	+3.32°	+3.06°	+0.26°
43 m	15.6°	+1.61°	+1.61°	0.00°
55 m	12.4°	+0.99°	+1.02°	–0.03°
70 m	9.9°	+0.61°	+0.64°	–0.03°
82 m	8.5°	+0.45°	+0.47°	–0.02°

It is within a third of a degree everywhere from 30 m out, and over-delivers only in the tightest turns — which is the normal behaviour of a steering trapezoid and harmless. **Tuning it to the median 43 m rather than to the extremes is the right choice**, and the reason the radius had to be measured before any of this could be specified.

The one input still to confirm — and it matters

L, the effective pivot distance, is the 12.6 m the boat *behaves* as though it has, derived from $R \cdot \tan \delta$. It is not a tape measurement, and the answer moves with it:

if L is	helm at 43 m	target	arm angle	ball joint inboard	tie rod
9.0 m	11.3°	+1.20°	24.1°	143 mm	4191 mm
10.5 m	13.1°	+1.38°	20.9°	125 mm	4227 mm
12.6 m	15.6°	+1.61°	17.5°	105 mm	4266 mm
14.0 m	17.2°	+1.76°	15.7°	95 mm	4287 mm
16.0 m	19.5°	+1.95°	13.6°	82 mm	4312 mm

From 9 m to 16 m the ball joint moves **143 mm to 82 mm** — a 60 mm span on a 349 mm arm, which is the difference between right and visibly wrong. The tie rod is far less sensitive, 4191 to 4312 mm.

So: measure the distance from the rudder stock to the centre of lateral resistance, and if it lands near 12.6 m, build the 105 mm. If it does not, the table above has the answer, and the reason for the disagreement is itself worth knowing.

One number that still does not fit

Working backwards from the measured behaviour, a symmetric tie-rod linkage would need its arms splayed **about 28° outboard** to produce the drift seen. That is enough splay to doubt the premise rather than the measurement.

Taken with the constant **–1.2°** offset — indistinguishable from one rudder sensor being zeroed differently — and an R^2 of only 0.05–0.14 on that drift, the likeliest reading is that **part of what looks like anti-Ackermann is instrumentation, not geometry**.

In order

1. **Zero both rudder sensors** dockside against physically aligned rudders and record the offsets. This alone may account for the **–1.2°**.
2. **Measure the tiller arms as they stand** — specifically how far inboard or outboard the ball joint currently sits from the stock. That single number says whether the linkage is the problem or the sensors are.
3. **Measure L**, stock to centre of lateral resistance, and check it against the 12.6 m the boat behaves as though it has.

4. **Then** move the ball joint to **105 mm inboard, 333 mm aft** and shorten the tie rod to **4266 mm** — adjusting per the L table if the tape disagrees.
5. **Re-measure.** A day on a 1–2 s log after the change, plotted against this same chart, either shows the differential has crossed to positive or it does not. Rate of turn and radius exist on only 8 days of 23 because the rest are logged at 6.09 s, so log the validation day finely.

Generated from `session_phase_stats` and the session logs, 26 September 2026. Linkage solved numerically as a symmetric steering trapezoid (two tiller arms, one tie rod, rudder separation 4.0 m), verified against the parallel-arm case which must and does return exactly parallel steering. Rate of turn: heading change over the 6 s centred on apparent wind = 0, on logs of 3 s or finer, gated on the crossing being interpolated across ≤ 3 s and the window holding ≥ 3 headings. Turn radius: mean boat speed over the same window divided by the yaw rate, above 2 °/s only. Linkage fit: ordinary least squares over log rows carrying both rudder angles with $|\text{steering}| \leq 15^\circ$. Effective pivot distance: $R \cdot \tan \delta$ with δ the mean helm over the turn, 2° – 25° .